

Quoridor – Python

Rapport préliminaire

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Introduction

Quoridor is an abstract strategy board game designed by Mirko Marchesi and first published in 1997 by the French company Gigamic. The game is played on a square grid and is based on two simple actions : moving a pawn and placing walls. Despite this apparent simplicity, the game exhibits a significant strategic depth, as players must constantly balance progress toward their own goal with the obstruction of their opponents.

The objective of this project is to design and implement a complete software version of the game Quoridor, supporting two, three, or four players. The program must strictly respect the official rules of the game and allow players to interact through both a command-line interface and a graphical user interface. In addition, the software must support configuration files, internationalization, and robust error handling, in accordance with the provided specifications.

This project goes beyond the development of a playable game. It aims to apply fundamental software engineering principles, including modular architecture, clear separation of concerns, automated testing, and comprehensive documentation. Quoridor also represents an interesting case study from an algorithmic perspective, as its gameplay can be naturally modeled using graph-based representations and shortest-path algorithms, which are commonly studied in computer science research.



FIGURE 1 – Figure 1 : Gigamic company (left) and Quoridor box (right)

1 Historical Background and Existing Work

Quoridor was invented by Mirko Marchesi in the mid-1990s and officially released in 1997 by the French publisher Gigamic. Since its publication, the game has gained international recognition for its elegant design and educational value, receiving several prestigious awards such as the Mensa Select Award in 1997 and the Games Magazine Game of the Year Award in 1998. Its accessibility, combined with its strategic depth, has contributed to its long-lasting popularity among both casual players and strategy game enthusiasts.

Beyond its commercial success, Quoridor has attracted attention in academic and technical contexts. The game's mechanics, which involve navigating a grid while dynamically placing obstacles, make it a natural candidate for algorithmic modeling. Several studies have explored Quoridor as a testbed for artificial intelligence, pathfinding algorithms, and adversarial search techniques.

In particular, previous academic work has shown that a Quoridor board can be represented as a graph in which vertices correspond to board squares and edges represent legal movements between adjacent squares. Wall placements dynamically modify this graph by removing edges, while pawn jumps introduce temporary alternative connections. Shortest-path algorithms such as breadth-first search or Dijkstra's algorithm are commonly used to evaluate positions, verify the legality of wall placements, and guide decision-making in artificial intelligence agents.

Notable contributions include undergraduate and master-level research that investigate heuristic evaluation functions, graph-based modeling, and search strategies for Quoridor. These works highlight the computational complexity of the game and demonstrate that, despite its simple rules, Quoridor poses non-trivial challenges for both human players and automated systems. This project builds upon these existing ideas while focusing primarily on the design of a robust, extensible, and well-engineered software implementation.

2 Première section

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3 Rules of the Game

This section describes the official rules of Quoridor as implemented in this project. The objective is to provide a precise and unambiguous description of the game mechanics so that a complete game can be played without interpretation issues.

3.1 Board and Components

Quoridor is played on a square board composed of a 9×9 grid, resulting in 81 playable squares. The squares are connected orthogonally, allowing movement in the four cardinal directions. Between these squares, grooves allow walls to be placed in order to block movement.

Each player controls a single pawn. In addition, a fixed number of walls is available and distributed among the players at the beginning of the game. Walls are barriers that block movement between adjacent squares and play a central role in shaping possible paths across the board.

3.2 Initial Setup

At the beginning of the game, the board is empty except for the players' pawns, which are placed at predefined starting positions.

In a two-player game, one pawn is placed at the center square of the bottom row, while the second pawn is placed at the center square of the top row. Each player's objective is to reach any square on the opposite side of the board.

In a four-player game, pawns are placed at the center of each side of the board : one on the top, one on the bottom, one on the left, and one on the right. Each player must reach the side opposite their starting position.

In the two-player configuration, each player receives ten walls. In the four-player configuration, each player receives five walls in order to maintain balance and avoid excessive blocking.

Players alternate turns, starting with one of the players chosen according to the game configuration.

3.3 Pawn Movement

On their turn, a player may move their pawn to an adjacent square. Movement is allowed only in the four orthogonal directions : forward, backward, left, or right. Diagonal movement is not allowed under normal circumstances.

A pawn may only move to an adjacent square if there is no wall blocking the path between the two squares. Pawns cannot move outside the boundaries of the board, nor can they move onto a square already occupied by another pawn.

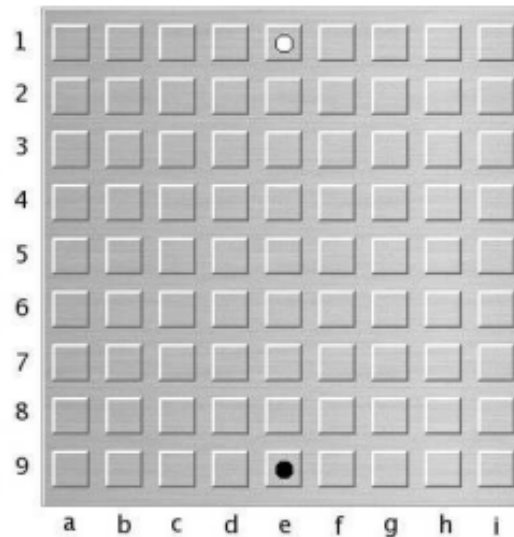


FIGURE 2 – Quoridor board layout

The goal of pawn movement is to progress toward the target edge of the board while navigating around walls placed by the opponents.

3.4 Jump Rules

If a pawn is located directly adjacent to an opponent's pawn and no wall separates them, the player may jump over the opponent's pawn. The jump moves the pawn to the square immediately beyond the opponent's pawn in the same direction.

If a direct jump is not possible because a wall blocks the square behind the opponent's pawn or because the board edge is reached, the player may instead move diagonally to one of the two adjacent squares to the left or right of the opponent's pawn, provided no wall blocks that diagonal movement.

In four-player games, a pawn may only jump over a single pawn per move. Multiple consecutive jumps over different pawns are not allowed.

These jump rules ensure that pawns cannot be completely blocked by the presence of other pawns and that movement remains possible even in crowded situations.

3.5 Wall Placement

Instead of moving their pawn, a player may choose to place a wall, provided they still have walls available. Walls are placed between squares and must be oriented either horizontally or vertically. Each wall spans exactly two adjacent square connections.

A wall placement must satisfy several constraints. Walls may not overlap or cross existing walls, and they must fit entirely within the board's grooves. Most importantly, a wall may not be placed if it completely blocks all possible paths for any player to reach their target side.

To ensure this rule is respected, at least one valid path from each pawn's current position to its goal edge must remain after the wall is placed. Pawn positions do not block these paths, as pawns can always be jumped over under the movement rules.

Once placed, walls are permanent and cannot be moved or removed for the remainder of the game.

3.6 Victory Condition

The game ends immediately when a player moves their pawn onto any square of their target edge. The first player to reach the opposite side of the board wins the game.

In the standard rules of Quoridor, draws are rare and generally not expected under normal play conditions, as players always have at least one legal move available until a pawn reaches its goal.

Voici un exemple de code en Python :

Un exemple d'image flottante (Fig. 5).

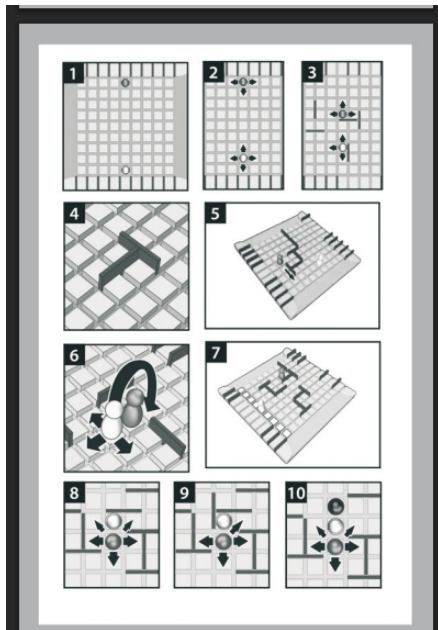


FIGURE 3 – Pawn movement

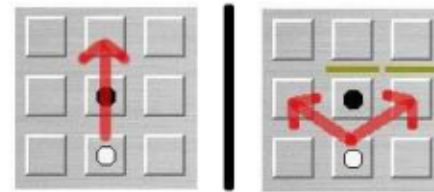


Figure 2.2: Allowed pawn jumps.



FIGURE 4 – Allowed jump

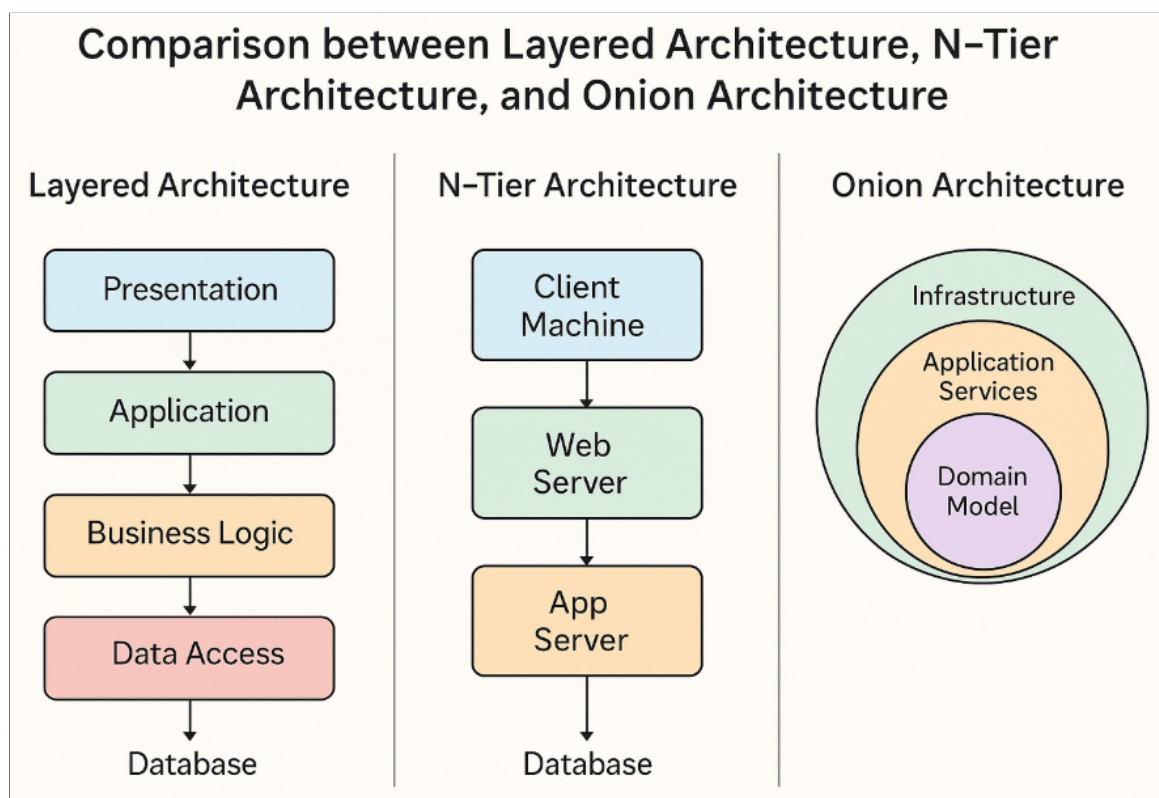


FIGURE 5 – Une architecture en couches

Conclusion

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A Section en annexe

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